

```
EOF'

sudo chown wwwrun /srv/www/cgi-bin/printenv
sudo chmod u+x /srv/www/cgi-bin/printenv

sudo systemctl start apache2

(c) For Ubuntu, use the following:

#!/bin/bash
#works for ubuntu gnu/linux
sudo timedatectl set-timezone 'America/Los_Angeles'
#timedatectl list-timezones
sudo apt-get update

#perl 5.18 comes installed by default but all required perl
modules, including SSL support, get installed with this
command
sudo apt-get -y install libapache2-mod-perl2

# Unlike other OS, apache web server needs to be separately
installed
# This will start the apache service as well
sudo apt-get -y install apache2

# Enable the CGI & Perl modules
cd /etc/apache2/mods-enabled
sudo ln -s ../mods-available/cgi.load
sudo ln -s ../mods-available/perl.load

sudo su - root -c 'cat >> /etc/apache2/conf-available/serve-
cgi-bin.conf << EOF

<IfModule mod_perl.c>

ScriptAlias /cgi-perl/ /usr/lib/cgi-bin/
<Location "/cgi-perl">
    AllowOverride None
    SetHandler perl-script
    PerlResponseHandler ModPerl::PerlRun
    PerlOptions +ParseHeaders
    Require all granted
</Location>

</IfModule>
EOF'

# Enable server software details to see the mod_perl and perl
versions
sudo sed -i '/^ServerTokens/c\ServerTokens Full' /etc/
apache2/conf-available/security.conf

sudo ln -sf /usr/bin/perl /usr/local/bin/perl # Create
symlink to default perl
```

```
# Creating the printenv file
sudo su - root -c 'cat > /usr/lib/cgi-bin/printenv << EOF
#!/usr/local/bin/perl

print "Content-type: text/plain; charset=iso-8859-1\n\n";
foreach $var (sort(keys(%ENV))) {
    print "$var=".$ENV{$var}."\n";
}
EOF'
sudo chown www-data /usr/lib/cgi-bin/printenv
sudo chmod u+x /usr/lib/cgi-bin/printenv
sudo service apache2 reload
```

- vi. Click *Next:Add Storage*. If you want, you can increase the storage space to 20. If you want more than that, then check the specifications allowed for free tier. But leave the *Volume* at just one *Root Volume*.
- vii. Click *Next:Add Tags*. You can leave this section unless you want to set a tag.
- viii. Click *Next:Configure Security Group*.
 - (a) Select the one that you had created in (2)(ii).
 - (b) Click *Review* and the *Launch*.
- ix. Click *Launch* on the next page, after we are satisfied with our settings.
- x. This will give a popup dialogue box in which you have to choose your key pair. If you had created just one key pair (not more), the one you created in (2)(i) will show up as selected, by default. Select the *Acknowledgement* and click *Launch Instances*.
- xi. In the resultant page, you will get a line that reads as follows: 'The following instance launches have been initiated:i-059ff3c697315eff8. View launch log. Click the first alphanumeric string, which will take you to your instance that is being launched.'
4. On the console now, towards the bottom, under the *Description* section, we can see the IP address to the right of the label — IPv4 Public IP. And to the right of the IP address, there is a small two-page icon, which is clickable. On clicking it, the IP address will get copied to the clipboard. Keep that copied to your notepad.
5. When the *Instance State* column says it is running, then you are ready to log in to your machine.
6. Step (3)(vi) can be done after logging into the instance or whichever the GNU/Linux machine we want to set up the Web environment on. We can copy-paste those lines into a *.sh* file, make it executable and run it. Log in to AWS (only required if using AWS).

What client to use

1. This section is required only if connecting to a remote machine and not if we are working on a local machine.
2. Command line SSH is built into Windows 10, MacOS and GNU/Linux. In MacOS, we just have to enable it separately. This includes all SSH client tools, e.g., ssh-keygen.